

EXPLAINABLE ARTIFICIAL INTELLIGENCE

L193 – Conclusion – Lent 2025



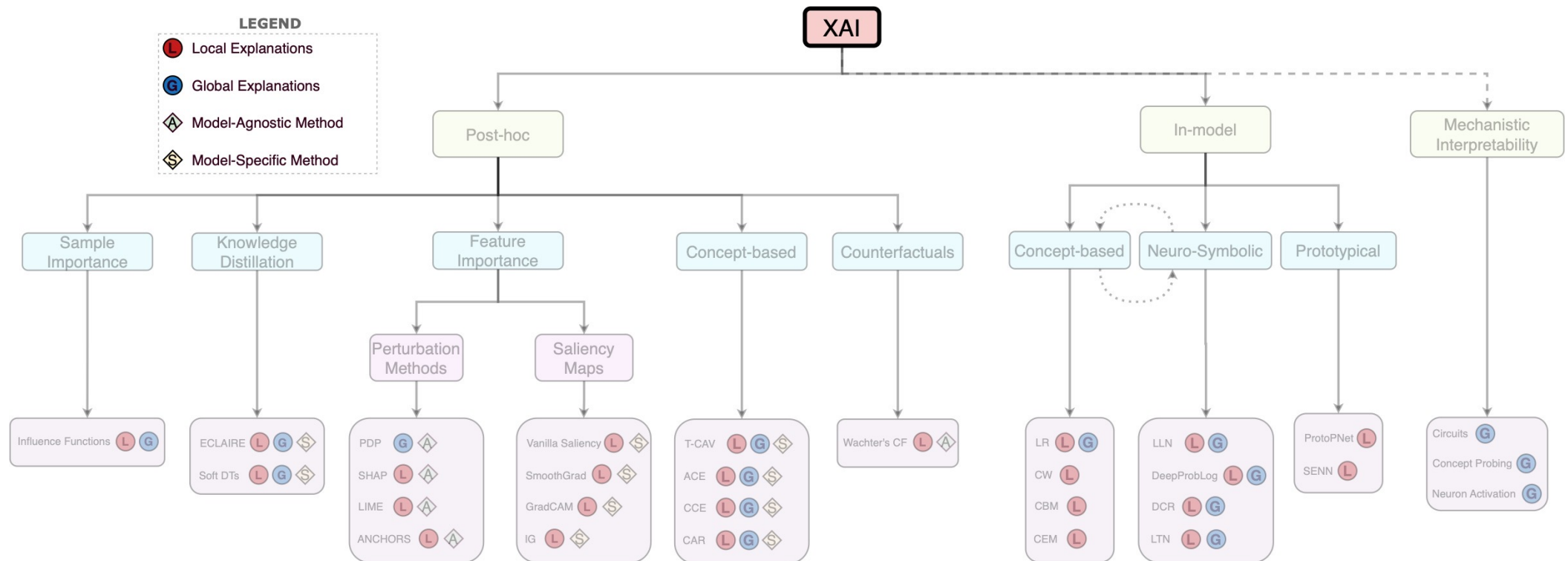
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THE ROAD AHEAD IN XAI

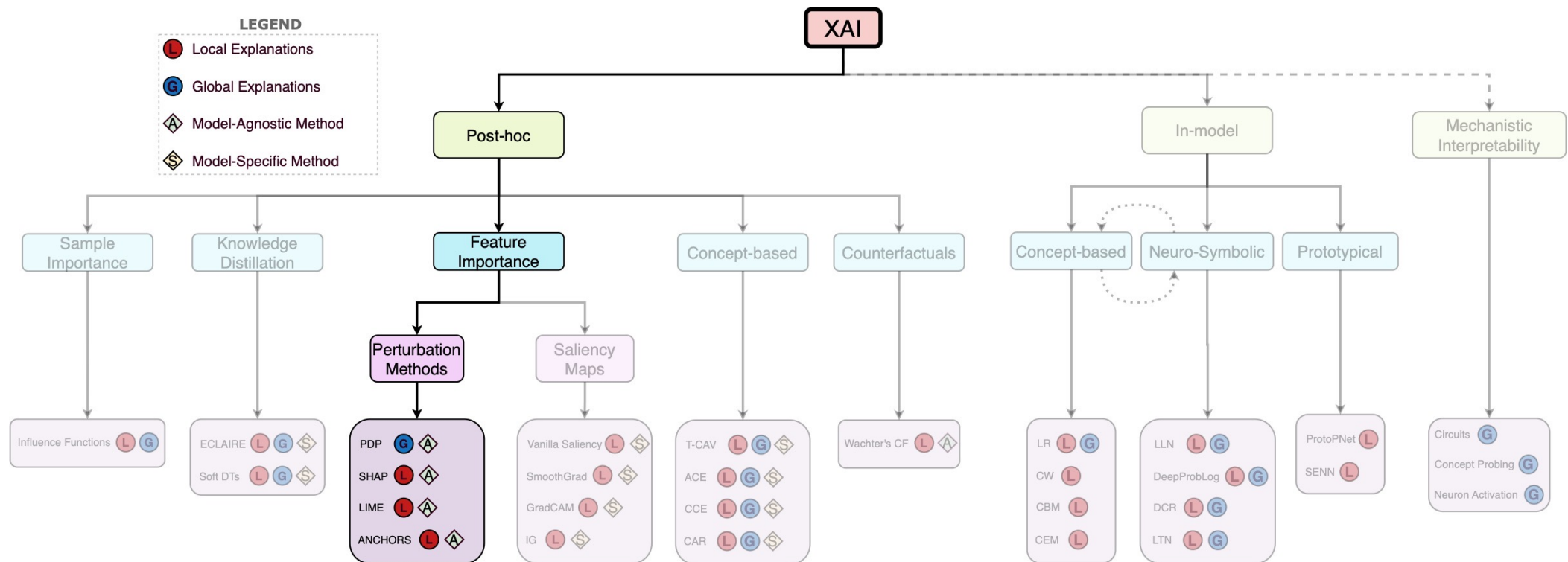


BEFORE ANYTHING, A LITTLE SUMMARY



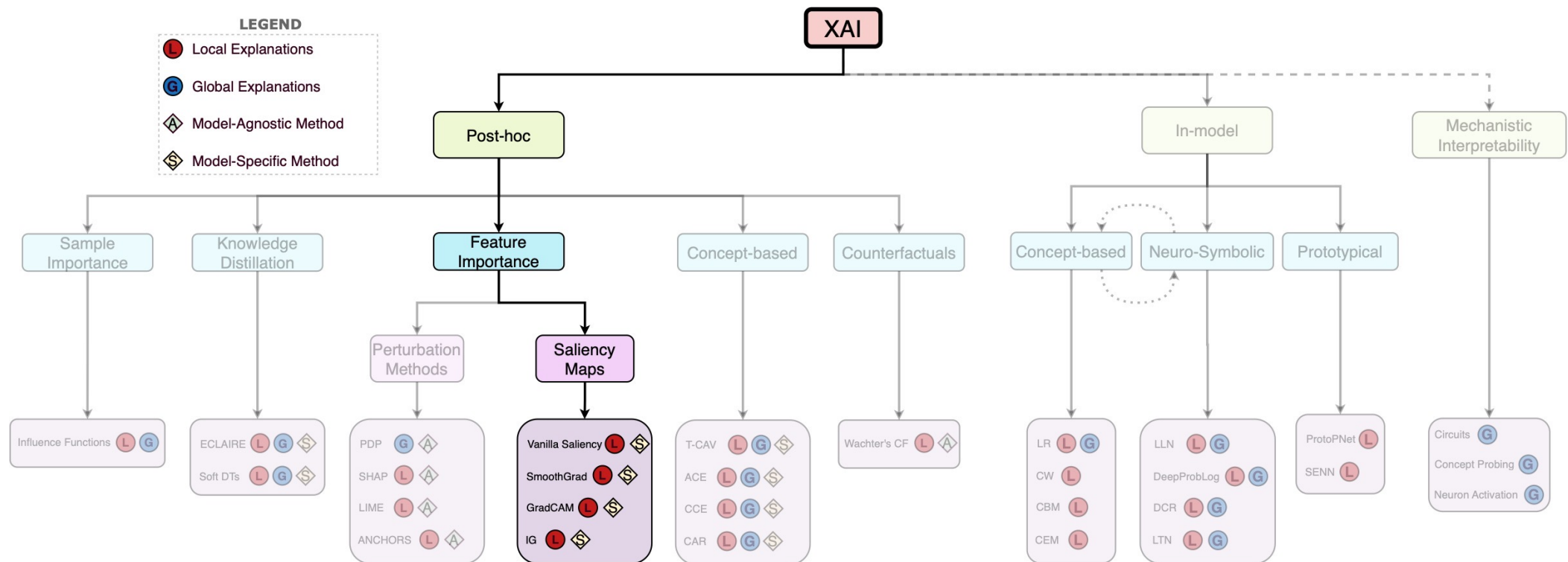
Let's take a step back and see the different areas of XAI that we discussed in the past eight weeks

BEFORE ANYTHING, A LITTLE SUMMARY



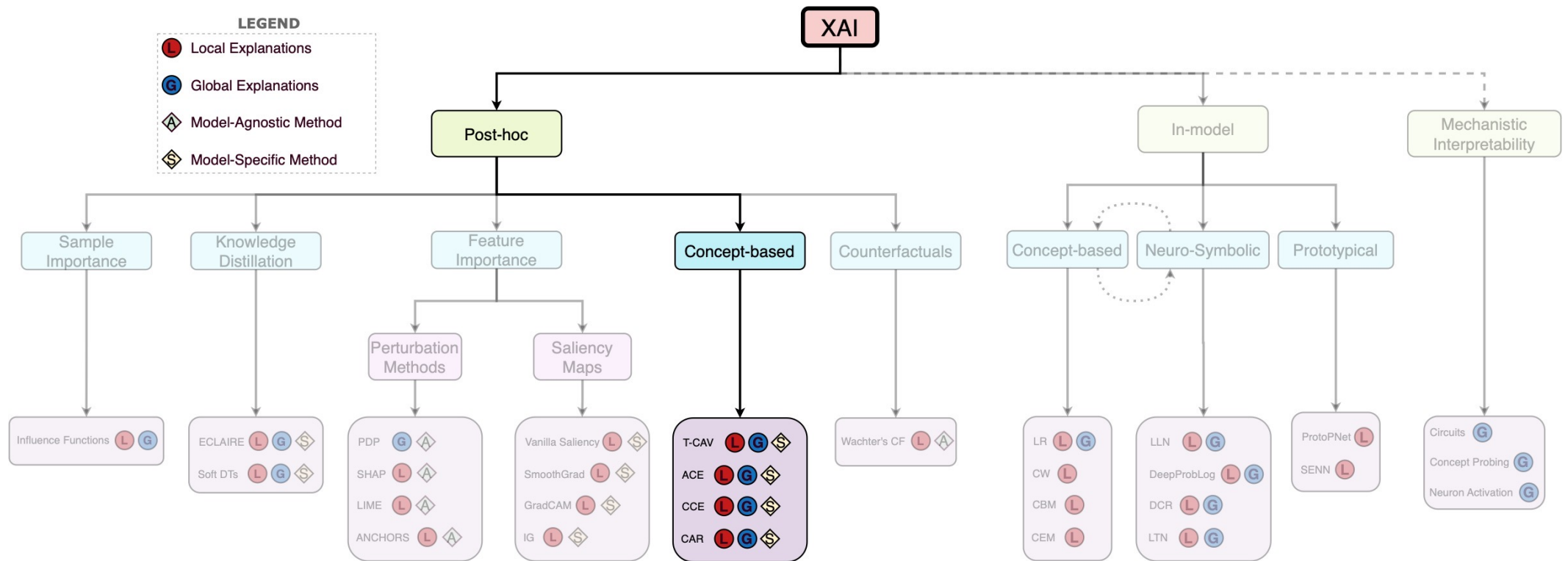
First, we discussed model-agnostic **feature importance** methods (so-called **perturbation methods**)

BEFORE ANYTHING, A LITTLE SUMMARY



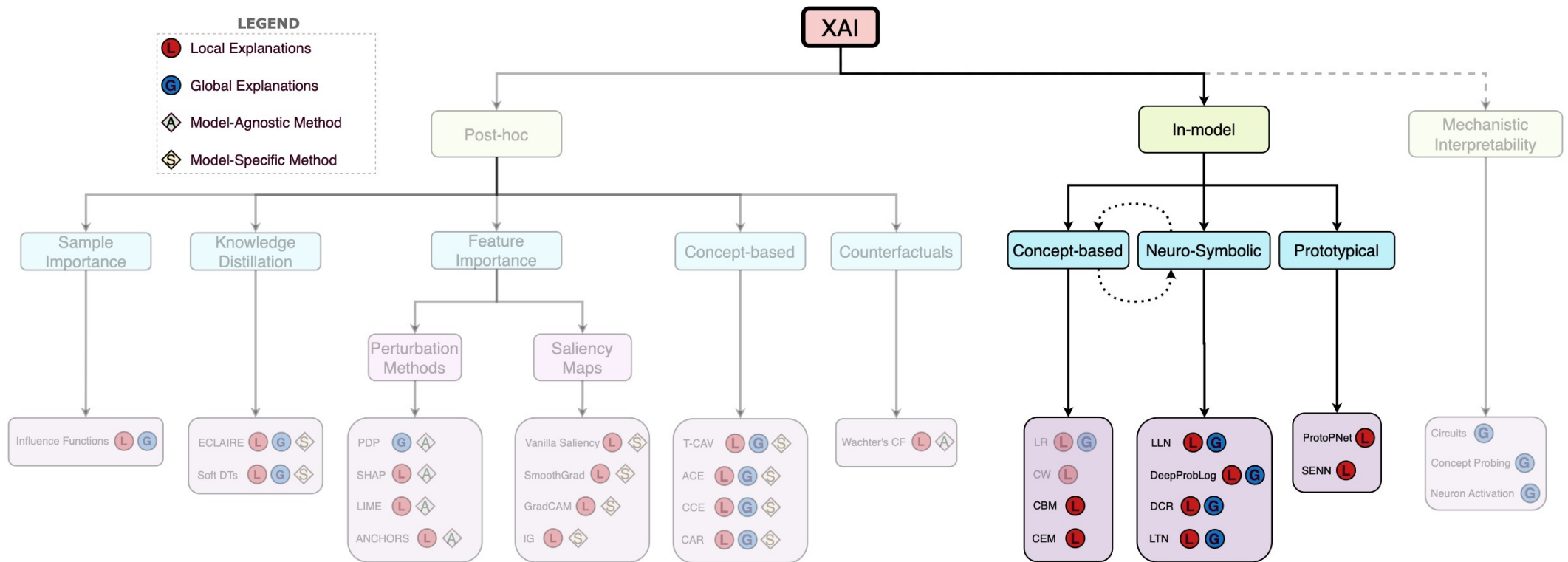
Then, we focused on model-specific **feature importance** methods like saliency maps

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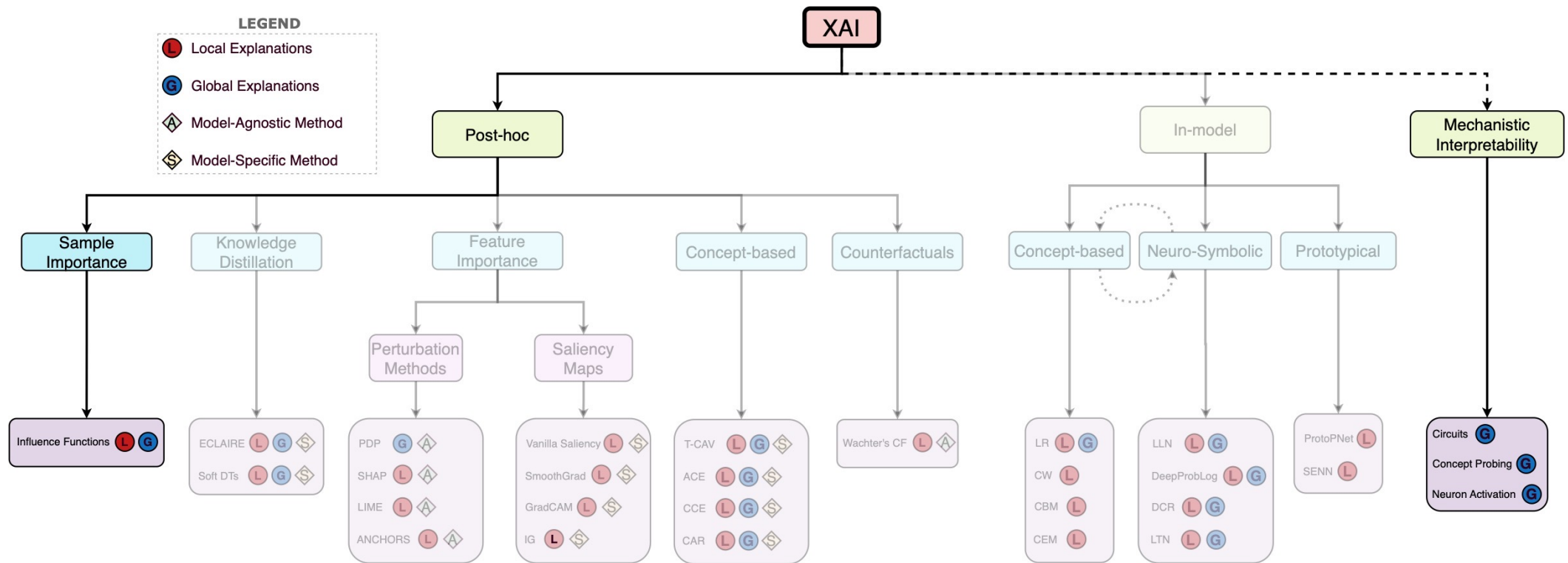
We followed up by exploring how to use **human-understandable concepts** to explain different models

BEFORE ANYTHING, A LITTLE SUMMARY



We discussed some limitations of post-hoc methods and introduced **in-model explainable methods**

BEFORE ANYTHING, A LITTLE SUMMARY



We concluded by briefly introducing recent trends such as **influence functions** and **mechanistic interpretability** 578

REMINISCING ABOUT THE FUTURE OF XAI



SOME REMINISCING ABOUT THE FUTURE OF XAI

With AI getting more and more intertwined with day-to-day activities, we conjecture that XAI will become a core component for:

1. **Regulation:** GDPR and the EU AI Act are just the beginning
2. **Standardisation:** most global efforts (telecommunications, internet, etc...) have been standardised under some international organisation, AI will soon follow
3. **Data science:** we will not analyse data, we will analyse models
4. **Scientific exploration:** Explaining and being explained is crucial for human understanding. Knowledge discovery can be exponentially improved if we are able to interrogate AIs

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Advancing mathematics by guiding human intuition with AI

[Alex Davies](#) ✉, [Petar Veličković](#), [Lars Buesing](#), [Sam Blackwell](#), [Daniel Zheng](#), [Nenad Tomašev](#), [Richard Tanburn](#), [Peter Battaglia](#), [Charles Blundell](#), [András Juhász](#), [Marc Lackenby](#), [Geordie Williamson](#), [Demis Hassabis](#) & [Pushmeet Kohli](#) ✉



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Thank
you!